Supplementary Material for the paper "LUNAR: Automated Testing of Multi-Turn Conversations in Task-Oriented LLM Systems"

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Table 1: System intents and their descriptions.

Intent	Description
`inform`	The system provides information relevant to the user's query without a follow-up question or call to action.
`inform_and_followup`	The system provides information and additionally poses a follow-up question or proposes an action.
`choice`	The system offers at least two concrete options to choose from, with no follow-up question.
`choice_and_followup`	The system offers at least two options and additionally poses a follow-up question.
`confirm`	The system confirms the requested action, without a follow-up question.
`confirm_and_followup`	The system confirms the requested action and additionally poses a follow-up question.
`clarify`	The system asks the user a clarifying question to resolve ambiguity.
`reject`	The system indicates it cannot fulfill the request, without a follow-up question.
`reject_and_followup`	The system indicates it cannot fulfill the request and poses a follow-up question or suggests an alternative.
`failure`	The system signals that something went wrong (e.g. it did not understand the input).
`misc`	The system's response does not fit any other category.

Table 2: (CarControl) Overview of features with categories and corresponding ranges. Null indicates that this featues is not used in the generation.

Feature	Category	Range
politeness	style	unpleasant < neutral < polite
anthropomorphism	style	very directive < directive < interrogative < empathic
slang	style	formal < neutral < rude
implicitness	style	not implicit < slightly implicit < implicit
word_perturbation	perturbation	{null, delete_words, introduce_fillers_llm}
system	content	{windows, fog_lights, ambient_lights, head_lights, climate, fan, seat_heating}
position	content	{null, driver, passenger, back_left, back_right}
seat_position	content	{null, driver, front_passenger, back_left, back_center, back_right}
fog_light_position	content	{null, front, rear}
window_state_target	content	{open, close}
onoff_state_target	content	{on, off}
head_lights_mode_target	content	{null, off, parking, low_beam, high_beam, auto}
climate_temperature_value_target	content	{null, 16–26}
seat_heating_level_target	content	{null, 0–3}
choice	intent	[0.001, 1.0]
add_preferences	intent	[0.001, 1.0]
ask	intent	[0.001, 1.0]
change_of_mind	intent	[0.001, 1.0]
confirmation	intent	[0.001, 1.0]
reject	intent	[0.001, 0.3]
reject_clarify	intent	[0.001, 0.5]
repeat	intent	[0.001, 0.5]
stop	intent	[0.001, 0.1]

Table 3: (Navi) Overview of features with categories and corresponding ranges.

Feature	Category	Range
category	content	{car_repair, restaurant, supermarket, bakery, bar, hotel, cinema, gas_station, hospital}
payment_method	content	{null, CASH, CREDIT_CARD, CONTACTLESS}
fuel_type	content	{null, SP95_E10, DIESEL, SP95, SP98}
food_type	content	{null, german, indian, italian, middle_eastern, french, chinese, japanese, thai, mexican, greek, vietnamese, turkish, american, vegan}
parking	content	{null, available}
word_perturbation	perturbation	{null, delete_words, introduce_fillers_llm, introduce_homophones_static}
rating	content	{null, 3.5, 4, 4.5}
fuel_price	content	{null, 1.5–1.9}
slang	style	formal < neutral < rude
implicitness	style	not implicit < slightly implicit < implicit
politeness	style	neutral < polite
anthropomorphism	style	very directive < directive < interrogative < empathic
choice	intent	[0.001, 1.0]
add_preferences	intent	[0.001, 1.0]
ask	intent	[0.001, 0.7]
change_of_mind	intent	[0.001, 1.0]
confirmation	intent	[0.99, 1.0]
reject	intent	[0.001, 0.3]
reject_clarify	intent	[0.001, 0.5]
repeat	intent	[0.001, 0.1]
stop	intent	[0.001, 0.1]

Table 4: Intent transition mapping: overview of permitted user intents which can follow system intents.

System Intent	Permitted Follow-up User Intents
`inform`	`add_preferences`, `ask`, `change_of_mind`, `stop`
`inform_and_followup`	`change_of_mind`, `add_preferences`, `confirmation`, `reject`, `reject_clarify`, `ask`, `stop`
`choice`	`choice`, `ask`, `add_preferences`, `change_of_mind`, `reject`, `reject_clarify`, `stop`
`choice_and_followup`	`choice`, `add_preferences`, `change_of_mind`, `reject`, `reject_clarify`, `stop`
`confirmation`	`change_of_mind`, `ask`, `add_preferences`, `stop`
`confirm_and_followup`	`change_of_mind`, `add_preferences`, `confirmation`, `stop`
`clarify`	`add_preferences`, `change_of_mind`, `ask`, `reject`, `stop`
`reject`	`change_of_mind`, `ask`, `stop`
`reject_and_followup`	`change_of_mind`, `confirmation`, `stop`
`failure`	`change_of_mind`, `repeat`, `stop`
`misc`	`change_of_mind`, `reject`, `reject_clarify`, `stop`

Table 5: Inter-rater and judge-human agreement for judge evaluation for Navi and CarControl use case. Interrate agreement for humans is calculated using fleiss kappa, while for judge-human agreement Cohens kappa is used.

		Request-orientation		Clarity	
Case Study	Model	α	κ	α	κ
Navi	DeepSeek-V3	0.88	0.68	0.82	0.59
	GPT-4o	0.83	0.58	0.48	0.30
	GPT-4o-mini	0.68	0.48	0.23	0.05
	GPT-5-mini	**0.90**	**0.79**	0.75	0.49
	Human†	0.81	0.66	0.64	0.39
Car Control	DeepSeek-V3	0.64	0.65	0.67	0.66
	GPT-4o	0.57	0.60	0.33	0.43
	GPT-4o-mini	0.57	0.50	0.19	0.316
	GPT-5-mini	**0.65**	**0.65**	0.72	0.71
	Human†	0.76	0.57	0.61	0.35

Table 6: Effectiveness (mean failures found) split by SUT model averaged over 6 runs.

Approach	LLM	CarControl-II	CarControl-I	Navi-II	Navi-I
LUNAR	GPT-4o	13.29	159.33	120.67	419.83
	GPT-5-Chat	13.50	85.56	137.00	314.57
LUNAR_S	GPT-4o	8.83	78.50	73.83	125.00
	GPT-5-Chat	14.83	24.33	114.00	84.00
SENSEI	GPT-4o	0.14	23.57	31.38	23.50
	GPT-5-Chat	0.00	7.67	11.33	17.17

Table 7: Failure rate (mean critical ratio) split by SUT model average over 6 runs.

Approach	LLM	CarControl-II	CarControl-I	Navi-II	Navi-I
LUNAR	GPT-4o	0.099	0.470	0.371	0.591
	GPT-5-Chat	0.141	0.326	0.292	0.597
LUNAR_S	GPT-4o	0.071	0.233	0.278	0.256
	GPT-5-Chat	0.171	0.124	0.278	0.329
SENSEI	GPT-4o	0.001	0.103	0.109	0.051
	GPT-5-Chat	0.000	0.042	0.047	0.051

Table 8: Efficiency results (time per failure, hours/failure) split by SUT model. Lower is better.

Approach	LLM	CarControl-II(2.0h)	CarControl-I (2.0h)	Navi-II (3.5h)	Navi-I (4.0h)
LUNAR	GPT-4o	0.1505	0.0126	0.0290	0.0095
	GPT-5-Chat	0.1481	0.0234	0.0255	0.0127
LUNAR_S	GPT-4o	0.2264	0.0255	0.0474	0.0320
	GPT-5-Chat	0.1348	0.0822	0.0307	0.0476
SENSEI	GPT-4o	14.0000	0.0849	0.1116	0.1702
	GPT-5-Chat	–	0.2609	0.3088	0.2330

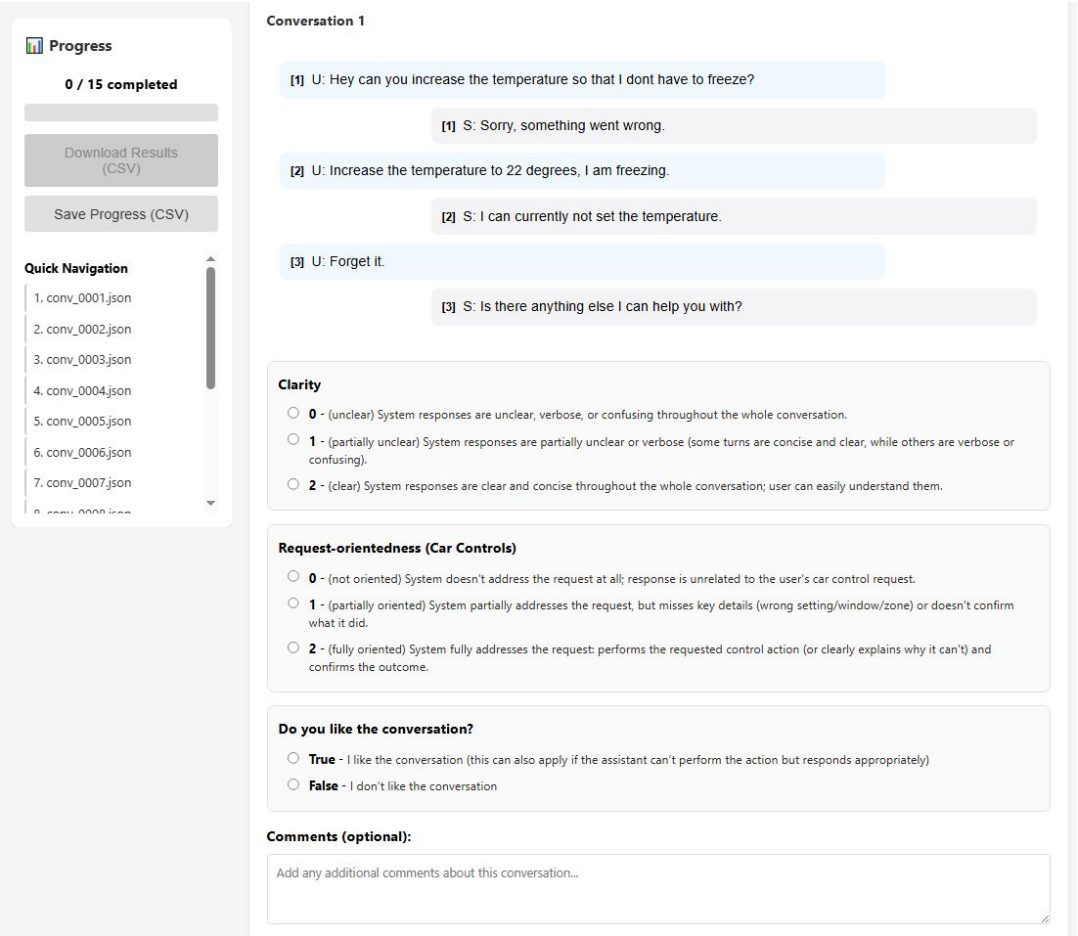

Figure 1: Overview of user interface for the survey for human-based data annotation for judge evaluation.

Table 9: (CarControl) Overview of features with categories and corresponding ranges. Null indicates that this featues is not used in the generation. Lowest intent probability is set to 0.001 to enable usage when no other intents are allowed to be chosen.

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